UIC Department of Chemical Engineering
University of Illinois at Chicago
COLLEGE OF ENGINEERING

ASSIGNMENT #04

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, March 20th TA's Mailbox

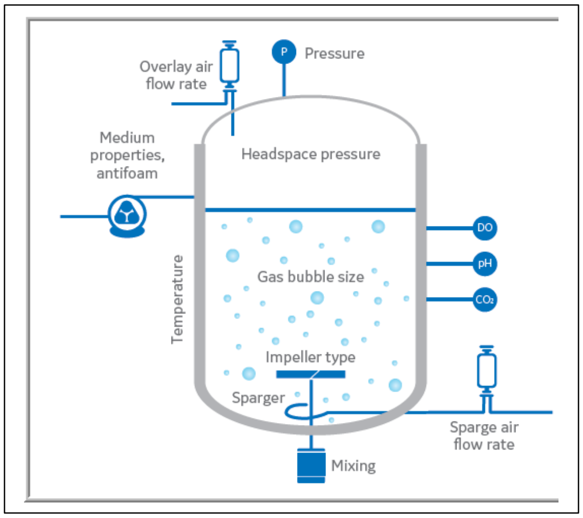
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Dissolved Oxygen (DO) Process Control

Air at atmospheric pressure is injected in the bottom of a bio-reactor as shown in the schematic.



The sparger creates bubbles in the reactor while the mixer reduces their size and facilitates the transport of oxygen from the bubbles to the liquid phase (broth) which is maintained at constant temperature equal to 35 °C. The mass transfer rate of oxygen from the bubbles to the broth is given by the Newton's law.

Oxygen is consumed in the liquid phase by the cells as they react with the substrate. The rate of consumption of oxygen is given by the equation: $R_{O_2} = K_{O_2} x$ where x is the cell concentration which is assumed to be constant. The following data are available for this biological process.

Parameter	Symbol	Units	Constant Value or Formula
Reactor volume	V	m^3	1.0
Cell concentration	x	$\mathrm{kg_{cell}/m^3}$	0.25
Reaction rate constant	K_{O_2}	$\mathrm{kmol/(kg_{cell} \cdot s)}$	1.1×10^{-4}
Air Flow Rate	F_{air}	$\mathrm{ft^3/min}$	500.0
Initial O_2 concentration in broth	$C_{O_2}^0$	$\mathrm{kmol/m^3}$	1.1×10^{-4}
Mass transfer coeffcient	$k_L\alpha$	1/s	$0.25+0.001(F_{air}-500)$

The flow rate of air in the mass transfer coefficient equation is in ft^3/min. The objective is to control the dissolved oxygen (DO) concentration in the broth by manipulating the air flow rate.

1. Develop a dynamic model for the concentration of oxygen in the broth. Next taking the Laplace transform of the dynamic model and performing the proper linearization and Skögestad approximations derive the FOPTD transfer function $G_P(s)$ of this process and calculate the process gain K_P, lag time τ, and time delay θ.
2. Perform a dynamic simulation until steady-state is reached. Then perturb the flow rate of air by 10% and from the response calculate the process gain, lag time and time delay.
3. Compare the values from your dynamic simulation with the values obtained by deriving the transfer function from the Laplace transform of the dynamic model and report the percent errors.

Assumptions and other pertinent data:

For the necessary physical properties, you may use ASPEN Plus to obtain the appropriate parameter values and you may assume that the air contains 79% Nitrogen and 21% Oxygen. You may also assume that the broth has the properties of water. The dissolved oxygen(DO) sensor has a lag time equal to $\tau_s = 30\,s$ and a repeatability of $\pm 0.5\%$

Chemical Process Control Report 4

Kyle Hankosky

March 2026

1 Dissolved Oxygen (DO) Process Control

1.1 Objective

The objective of this problem is to develop a dynamic model for dissolved oxygen concentration in a bioreactor, linearize the model about the nominal operating condition, and derive a first-order plus time delay (FOPTD) transfer function for the process. The air flow rate is the manipulated variable and the dissolved oxygen concentration in the broth is the controlled variable.

1.2 Given Data

Parameter	**Units**	**Value**
Reactor volume, V	m^3	1.0
Cell concentration, x	$\text{kg}_{cell}/\text{m}^3$	0.25
Reaction rate constant, K_{O_2}	$\text{kmol}/(\text{kg}_{cell}{\cdot}\text{s})$	1.1×10^{-4}
Nominal air flow rate, $F_{air,s}$	ft^3/min	500
Initial dissolved oxygen concentration, $C^0_{O_2}$	kmol/m^3	1.1×10^{-4}
Mass transfer coefficient correlation, $k_L a$	1/s	$0.25 + 0.001(F_{air} - 500)$
Sensor lag time, τ_s	s	30
Equilibrium DO concentration, C^*	kmol/m^3	2.19×10^{-4}

Table 1: Process data used for the dissolved oxygen control model.

1.3 Dynamic Model Development

A transient mass balance on dissolved oxygen in the liquid phase gives

$$\text{Accumulation} = \text{Oxygen transfer from gas bubbles} - \text{Oxygen consumption by cells}$$

The oxygen transfer rate is modeled by Newton's law:

$$r_{\text{transfer}} = k_L a (C^* - C)$$

The oxygen consumption rate is given in the problem statement as

$$R_{O_2} = K_{O_2} x$$

Therefore, the dynamic mass balance becomes

$$V\frac{dC}{dt} = V\,k_L a(C^* - C) - V\,K_{O_2}x$$

Since the reactor volume is $V = 1.0$ m^3, the model simplifies to

$$\frac{dC}{dt} = k_L a(C^* - C) - K_{O_2}x$$

Substituting the given correlation for the mass transfer coefficient,

$$k_L a = 0.25 + 0.001(F_{air} - 500)$$

the nonlinear dynamic model is

$$\boxed{\frac{dC}{dt} = [0.25 + 0.001(F_{air} - 500)]\,(C^* - C) - K_{O_2}x}$$

1.4 Steady-State Calculation at the Nominal Operating Point

At the nominal air flow rate,

$$F_{air,s} = 500 \text{ ft}^3/\text{min}$$

the mass transfer coefficient is

$$k_L a_s = 0.25 + 0.001(500 - 500) = 0.25 \text{ s}^{-1}$$

The oxygen consumption term is

$$K_{O_2}x = (1.1 \times 10^{-4})(0.25) = 2.75 \times 10^{-5} \text{ kmol/(m}^3\text{s)}$$

At steady state,

$$\frac{dC}{dt} = 0$$

so

$$0 = k_L a_s(C^* - C_s) - K_{O_2}x$$

Solving for the steady-state dissolved oxygen concentration,

$$C_s = C^* - \frac{K_{O_2}x}{k_L a_s}$$

Substituting values,

$$C_s = 2.19 \times 10^{-4} - \frac{2.75 \times 10^{-5}}{0.25}$$

$$C_s = 2.19 \times 10^{-4} - 1.10 \times 10^{-4}$$

$$\boxed{C_s = 1.09 \times 10^{-4} \text{ kmol/m}^3}$$

This value is consistent with the stated initial dissolved oxygen concentration of approximately 1.1×10^{-4} kmol/m^3.

1.5 Linearization About Steady State

Define deviation variables

$$C' = C - C_s \qquad \text{and} \qquad F' = F_{air} - F_{air,s}$$

The nonlinear model can be written as

$$\frac{dC}{dt} = f(C, F_{air}) = k(F_{air})(C^* - C) - K_{O_2}x$$

where

$$k(F_{air}) = 0.25 + 0.001(F_{air} - 500)$$

Linearizing about the steady state gives

$$\frac{dC'}{dt} = \left.\frac{\partial f}{\partial C}\right|_s C' + \left.\frac{\partial f}{\partial F_{air}}\right|_s F'$$

The partial derivative with respect to concentration is

$$\left.\frac{\partial f}{\partial C}\right|_s = -k_s = -0.25$$

The partial derivative with respect to air flow rate is

$$\frac{\partial f}{\partial F_{air}} = \frac{dk}{dF_{air}}(C^* - C)$$

Since

$$\frac{dk}{dF_{air}} = 0.001$$

then at steady state,

$$\left.\frac{\partial f}{\partial F_{air}}\right|_s = 0.001(C^* - C_s)$$

Using

$$C^* - C_s = \frac{K_{O_2}x}{k_s} = \frac{2.75 \times 10^{-5}}{0.25} = 1.10 \times 10^{-4}$$

gives

$$\left.\frac{\partial f}{\partial F_{air}}\right|_s = 0.001(1.10 \times 10^{-4}) = 1.10 \times 10^{-7}$$

Thus, the linearized model is

$$\boxed{\frac{dC'}{dt} = -0.25C' + 1.10 \times 10^{-7}F'}$$

1.6 Laplace Transform and Process Transfer Function

Taking the Laplace transform of the linearized model with zero initial conditions,

$$sC'(s) = -0.25C'(s) + 1.10 \times 10^{-7}F'(s)$$

Rearranging,

$$(s + 0.25)C'(s) = 1.10 \times 10^{-7}F'(s)$$

so the transfer function from air flow rate to dissolved oxygen concentration is

$$\frac{C'(s)}{F'(s)} = \frac{1.10 \times 10^{-7}}{s + 0.25}$$

Writing this in standard first-order form,

$$\frac{C'(s)}{F'(s)} = \frac{4.40 \times 10^{-7}}{4s + 1}$$

Therefore, the process transfer function is

$$\boxed{G_p(s) = \frac{4.40 \times 10^{-7}}{4s + 1}}$$

From this expression, the process parameters are

$$\boxed{K_p = 4.40 \times 10^{-7}\ \frac{\text{kmol/m}^3}{\text{ft}^3\text{/min}}}$$

$$\boxed{\tau_p = 4\ \text{s}}$$

$$\boxed{\theta_p = 0\ \text{s}}$$

1.7 Inclusion of Sensor Lag

The problem states that the dissolved oxygen sensor has a first-order lag time of 30 s. The corresponding sensor transfer function is

$$G_s(s) = \frac{1}{30s + 1}$$

The overall measured response is therefore

$$G_{\text{measured}}(s) = G_p(s)G_s(s) = \frac{4.40 \times 10^{-7}}{(4s + 1)(30s + 1)}$$

1.8 FOPTD Approximation

To express the measured process in FOPTD form, Skogestad's approximation was applied. The dominant lag is the 30 s sensor lag, while the smaller 4 s process lag is approximated as half added to the retained time constant and half added to the effective dead time. Therefore,

$$\tau = 30 + \frac{4}{2} = 32 \text{ s}$$

$$\theta = \frac{4}{2} = 2 \text{ s}$$

Thus, the final FOPTD approximation is

$$\boxed{G_{FOPTD}(s) = \frac{4.40 \times 10^{-7} e^{-2s}}{32s + 1}}$$

1.9 Results

Quantity	**Units**	**Value**
Steady-state DO concentration, C_s	kmol/m^3	1.09×10^{-4}
Process gain, K_p	$(\text{kmol/m}^3)/(\text{ft}^3/\text{min})$	4.40×10^{-7}
Process time constant, τ_p	s	4.0
Process dead time, θ_p	s	0.0
FOPTD time constant, τ	s	32.0
FOPTD dead time, θ	s	2.0

Table 2: Derived process and FOPTD model parameters for dissolved oxygen control.

2 Dynamic Simulation and Step Disturbance Response

A dynamic simulation was performed using the nonlinear time-domain model developed in Question 1. The simulation was carried out using the CHE Calculator ODE solver with the Runge-Kutta 5th order (RK5) integration method. The simulation was divided into two stages. In the first stage, the system was simulated at the nominal air flow rate of 500 ft^3/min until the dissolved oxygen concentration and the sensor output approached steady state. In the second stage, a 10% step increase in air flow rate was applied, changing the input from 500 to 550 ft^3/min, and the resulting response of the dissolved oxygen sensor was used to estimate the process gain, lag time, and time delay.

The governing equations used in the simulation were

$$\frac{dC}{dt} = [0.25 + 0.001(F_{air} - 500)]\,(C^* - C) - K_{O_2}x$$

$$\frac{dC_m}{dt} = \frac{C - C_m}{30}$$

where C is the actual dissolved oxygen concentration in the broth and C_m is the measured dissolved oxygen concentration from the sensor. The initial conditions used for the simulation were

$$C(0) = 1.10 \times 10^{-4}\ \text{kmol/m}^3$$

$$C_m(0) = 0$$

The sensor output was allowed to settle prior to the disturbance so that the step-response analysis could be performed on the post-disturbance data only.

2.1 Simulation Conditions

The numerical integration settings used in the CHE Calculator ODE solver are summarized below.

Parameter	Value
Integration method	RK5 (Runge-Kutta 5th order)
Time range	0 – 300 s
Step change time	150 s
Number of solution points	20
Steps between solution points	10
Intermediate breakpoints	0.5
Relative tolerance	1×10^{-6}
Absolute tolerance	1×10^{-6}
Number of equations	2

Table 3: ODE solver settings used for the dynamic simulation.

The process conditions used in the simulation are summarized below.

Parameter	Units	Value
Initial air flow rate, $F_{air,1}$	ft^3/min	500
Disturbed air flow rate, $F_{air,2}$	ft^3/min	550
Input step size, ΔF_{air}	ft^3/min	50
Initial broth DO concentration, $C(0)$	kmol/m^3	1.10×10^{-4}
Initial sensor reading, $C_m(0)$	kmol/m^3	0
Sensor time constant, τ_s	s	30

Table 4: Process conditions used for the dissolved oxygen step test.

2.2 Simulation Results

From the simulation, the measured dissolved oxygen concentration immediately before the disturbance was

$$y_0 = 1.08267 \times 10^{-4} \text{ kmol/m}^3$$

The response increased after the 10% air-flow step and approached a practical final value of

$$y_f = 1.27016 \times 10^{-4} \text{ kmol/m}^3$$

The process gain was determined directly from the steady-state change in output divided by the input step size:

$$K_p = \frac{y_f - y_0}{\Delta F_{air}}$$

$$\boxed{K_p = 3.74996 \times 10^{-7} \ \frac{\text{kmol/m}^3}{\text{ft}^3/\text{min}}}$$

2.3 Determination of τ and θ

The lag time and time delay were estimated using the 28.3% and 63.2% step-response method. The target response values were calculated as

$$y_{28} = y_0 + 0.283(y_f - y_0)$$

$$y_{63} = y_0 + 0.632(y_f - y_0)$$

$$\boxed{y_{28} = 1.13573 \times 10^{-4} \text{ kmol/m}^3} \qquad \boxed{y_{63} = 1.20116 \times 10^{-4} \text{ kmol/m}^3}$$

Using linear interpolation of the simulated data,

$$\boxed{T_{28} = 163.243 \text{ s}} \qquad \boxed{T_{63} = 182.666 \text{ s}}$$

The time constant was calculated as

$$\tau = 1.5(T_{63} - T_{28}) = 29.13 \text{ s}$$

$$\boxed{\tau = 29.13 \text{ s}}$$

The absolute dead time was calculated as

$$\theta = T_{63} - \tau = 153.53 \text{ s}$$

Since the disturbance occurred at 150 s, the relative dead time is

$$\boxed{\theta = 3.53 \text{ s}}$$

The resulting FOPTD model from the simulation is

$$\boxed{G_p(s) = \frac{3.75 \times 10^{-7} e^{-3.53s}}{29.13s + 1}}$$

2.4 Step Response Plot

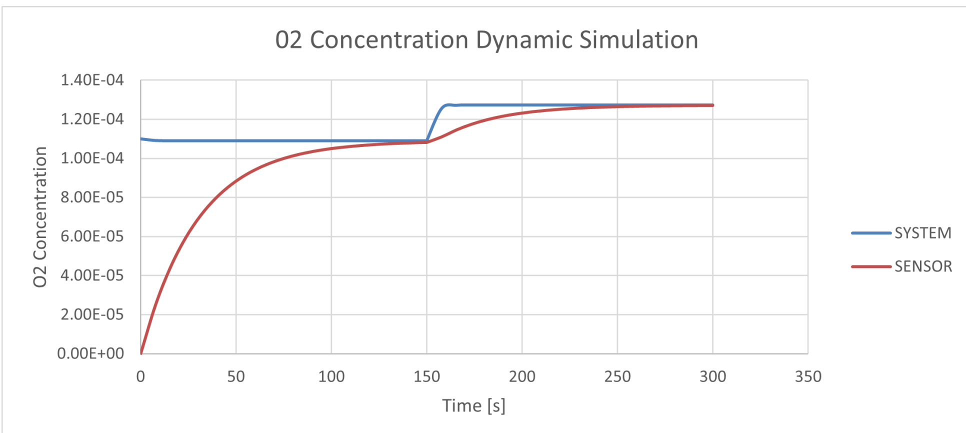


Figure 1: Dynamic simulation of dissolved oxygen concentration for a 10% step increase in air flow rate.

2.5 Results Summary

Quantity	**Units**	**Value**
Process gain, K_p	$(\text{kmol/m}^3)/(\text{ft}^3/\text{min})$	3.75×10^{-7}
Time constant, τ	s	29.13
Dead time, θ	s	3.53

Table 5: FOPTD parameters obtained from dynamic simulation.

3 Comparison of Derived Model and Dynamic Simulation

The FOPTD parameters obtained from the analytical model in Question 1 were compared to those obtained from the dynamic simulation in Question 2. The purpose of this comparison is to evaluate how well the Laplace-derived model captures the true dynamic behavior of the system.

The parameters from the derived model (Q1) are

$$K_{p,\text{Q1}} = 4.40 \times 10^{-7} \qquad \tau_{\text{Q1}} = 32 \text{ s} \qquad \theta_{\text{Q1}} = 2.0 \text{ s}$$

The parameters obtained from the dynamic simulation (Q2) are

$$K_{p,\text{Q2}} = 3.74996 \times 10^{-7} \qquad \tau_{\text{Q2}} = 29.13 \text{ s} \qquad \theta_{\text{Q2}} = 3.53 \text{ s}$$

Percent error was calculated using

$$\% \text{ error} = \left| \frac{\text{simulation} - \text{derived}}{\text{derived}} \right| \times 100$$

3.0.1 Process Gain Error

$$\% \text{ error in } K_p = \left| \frac{3.74996 \times 10^{-7} - 4.40 \times 10^{-7}}{4.40 \times 10^{-7}} \right| \times 100$$

$$\boxed{\% \text{ error in } K_p = 14.77\%}$$

3.0.2 Time Constant Error

$$\% \text{ error in } \tau = \left| \frac{29.13 - 32}{32} \right| \times 100$$

$$\boxed{\% \text{ error in } \tau = 8.95\%}$$

3.0.3 Dead Time Error

$$\% \text{ error in } \theta = \left| \frac{3.53 - 2.0}{2.0} \right| \times 100$$

$$\boxed{\% \text{ error in } \theta = 76.56\%}$$

3.1 Results Summary

Parameter	**Derived Model (Q1)**	**Simulation (Q2)**	**Percent Error**
K_p	4.40×10^{-7}	3.74996×10^{-7}	14.77%
τ (s)	32.0	29.13	8.95%
θ (s)	2.0	3.53	76.56%

Table 6: Comparison of FOPTD parameters from analytical model and dynamic simulation.

3.2 Discussion

The derived model and the dynamic simulation gave similar values for process gain and time constant, so the overall system behavior was predicted reasonably well. The largest difference was in the dead time, which was expected since the delay was very small and more sensitive to approximation. Overall, the transfer function from Question 1 provided a good estimate of the process response observed in Question 2.

Use of External Resources

This report was completed using course materials, and ChatGPT was used as a support tool to assist with understanding concepts, verifying calculations, and formatting the final report in LaTeX.